1. Toolkit

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q} \ (|q|<1) \qquad \sum_{n=1}^{\infty} q^n = \frac{q}{1-q}$$
$$\sum_{n=0}^{N-1} q^n = \frac{1-q^N}{1-q} \ (q\neq 1)$$

- **Geometric: first term/(1 − ratio)**, |ratio| < 1. $\Sigma_{a..b} q^n = (q^a - q^{b+1})/(1-q)$. $\Sigma_{n\geq 0}(az^{-1})^n = z/(z-a)$ if $|z| > |a|$; $\Sigma_{n\geq 1}(z/a)^n = (z/a)/(1-z/a)$ if $|z| < |a|$; $z = e^{j\Omega}$ → DTFT.
- **Euler:** $\cos\theta = (e^{j\theta} + e^{-j\theta})/2$, $\sin\theta = (e^{j\theta} - e^{-j\theta})/(2j)$. $1/j = -j$, $e^{j\pi/2} = j$, $e^{j\pi} = -1$.
- **Sampling:** $x[n] = x_c(nT_s)$, $T_s = 1/f_s$. $\Omega = \omega T_s = 2\pi f/f_s$ (rad/sample).
- **Period of cos(Ω_0n):** write $\Omega_0/2\pi = m/N$ in lowest terms → $N_0 = N$ (the denominator).
  - $\sin(\pi n/12)$: 1/24 → $N_0 = 24$. $\cos(3.5\pi n)$: 7/4 → $N_0 = 4$.
  - $\Omega_0/2\pi$ irrational (e.g. cos(n)) → not periodic.
- **Sum of periodic signals:** $N_0 = \text{LCM}(N_1, N_2)$.
- **Nonzero periodic signal is never absolutely summable** ($\Sigma|x|$ adds the same block forever).
- **Aliasing:** Ω, Ω + 2πk and (for cos) −Ω give identical samples. In Hz: f, f + k·f_s, f_s − f look the same.
  - e.g. $\cos(3.5\pi n) = \cos(-0.5\pi n) = \cos(0.5\pi n)$.
  - Unique only if $0 \leq \Omega \leq \pi$, i.e. $f < f_s/2$. Nyquist: $f_s > 2f_{max}$.

2. Systems and Convolution

Property	Means	Quick test / red flags
Linear	T(ax+bz) = a·Tx + b·Tz	Double the input: does the output double? Fails with x^2, x[n]·x[n−1], + constant
Time-inv.	input shifted by r → output shifted by r	Fails if n appears outside x[·]: n·x[n], x[2n], x[−n]
Causal	y[n] uses only x[n], x[n−1], …	Fails with any x[n+1] (future input)
BIBO stable	$\|x\| \leq M \rightarrow \|y\| \leq C$	Bound term by term: $\|x[n]+x[n-1]+x[n]x[n-1]\| \leq 2M + M^2$

LTI: $h = T(\delta)$; $y = x * h$, $(x*h)[n] = \Sigma_m x[m]\cdot h[n-m]$

- **LTI causal** ⇔ h[n] = 0 for n < 0. **LTI stable** ⇔ $\Sigma|h[n]| < \infty$.
- **Convolution facts:** $x * \delta[n-k] = x[n-k]$. $x * u$ = running sum $\Sigma_{m\leq n} x[m]$. $u * u = (n+1)u[n]$ ($\Sigma_m u[m]u[n-m]$ counts m = 0…n → n+1).
- **Step → impulse response:** s = T(u) ⇒ h[n] = s[n] − s[n−1] (because $\delta[n] = u[n] - u[n-1]$).
- **Length:** linear conv of L_x and L_h samples has $L_x + L_h - 1$ samples.
- **FIR:** no y on the right; h = the coefficients; all poles at z = 0; always stable.
- **IIR:** feedback (y[n−k] on the right); h never ends; stable only if all poles inside |z| = 1.

3. DFT and DTFS

analysis $X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}$

synthesis $x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{+j2\pi kn/N}$

- **Bin k** ↔ $\Omega_k = 2\pi k/N$ ↔ $f_k = k\cdot f_s/N$.
- **N = 4:** X = (matrix)·x. Synthesis x = ¼·(same matrix with j ↔ −j)·X.

Analysis row	n=0	n=1	n=2	n=3	Synthesis row (×¼)
k=0	1	1	1	1	1, 1, 1, 1
k=1	1	−j	−1	j	1, j, −1, −j
k=2	1	−1	1	−1	1, −1, 1, −1
k=3	1	j	−1	−j	1, −j, −1, j

- **Quick values:** X[0] = Σx[n]. x[0] = (1/N)ΣX[k]. N = 4: $X[2] = x_0 - x_1 + x_2 - x_3$.
- **Real x:** X[N−k] = X[k]* (N = 4: X[3] = X[1]*); X[0] and X[N/2] are real. Rows 1 and 3 are conjugates.
- **Unknowns in x and X (T02 Q1d style):** fill in X[3] = X[1]* first, then use one synthesis row for a known x value.
- **DTFS:** same formulas, valid for all n only if N is a period (multiple of N_0); otherwise the repeated block ≠ the signal (leakage).
- **N_0 vs N:** N_0 = signal's period (calculate). N = DFT size (choose). No leakage ⇔ N is a multiple of N_0.
- **DTFS by Euler (no sum):** write x as a sum of $e^{j2\pi kn/N}$; then X[k] = N × (coefficient of $e^{j2\pi kn/N}$). Use $e^{-j2\pi n/N} = e^{j2\pi(N-1)n/N}$.
  - e.g. $\cos(2\pi n/8) = \frac{1}{2}e^{j2\pi n/8} + \frac{1}{2}e^{j2\pi\cdot 7n/8}$ → X[1] = X[7] = 8 × ½ = 4.
  - For sin: $1/(2j) = \frac{1}{2}e^{-j\pi/2}$.

4a. Circular Convolution

- **Multiplying DFTs gives circular convolution:** $\text{IDFT}_N\{X[k]\cdot Y[k]\} = x \circledast_N y$, not the linear one.
- **Wrap rule:** C[n] = L[n] + L[n+N] (+ L[n+2N] …): fold the tail of the linear result L back onto its start.
  - e.g. L = [−2, 1, 3, −6, 2, 8], N = 4 → C = [−2+2, 1+8, 3, −6] = [0, 9, 3, −6].
- **Circular = linear:** zero-pad both to $N \geq L_x + L_y - 1$.

R. Question-type Recipes (T01–T02)

On the sheet already: 1 Period N_0 *T01 Q1* → §9 *Simplify first*, then §1 m/N rule · **2 Same samples** *T01 Q2* → §1 *Aliasing* · **3 Properties** *T01 Q3* → §2 table + §9 *Causal*, *Linearity* · **4 Step → h** *T01 Q4* → §2: h[n] = s[n] − s[n−1], then test h (finite-length h ⇒ stable) · **5 4-pt DFT matrix** *T02 Q1a* → §3 table (entry (k,n) = $(-j)^{kn}$, kn mod 4) · **9 Linear vs circular** *T02 Q4* → §4a.

6 Missing x / X values *T02 Q1d*: X[N−k] = X[k]* fills X → one synthesis row at a known sample, e.g. $4v[2] = V_0 - V_1 + V_2 - V_3$ → synthesise the rest.

7 DTFS without the sum *T02 Q2b*: (i) N_0 → (ii) Euler-split into $e^{j2\pi kn/N}$ terms (or sin → cos(θ−π/2) first) → (iii) rename negative k → N−k → (iv) X[k] = N × weight → (v) check at n = 0.

8 N-pt synthesis = x? *T02 Q2c*: only if N is a multiple of N_0; else exact on 0…N−1 only (check x[N] vs x[0]).

10 DTFT of an exponential *T02 Q3*: (i) exists ⇔ $\Sigma|x| < \infty$ ⇔ |ratio| < 1 ($a^{-n}u[n]$: $|a| > 1$) → (ii) $x[n]e^{-j\Omega n} = q^n$, $q = \text{ratio}\cdot e^{-j\Omega}$, $|q| = |\text{ratio}| < 1$ → (iii) $\Sigma q^n = 1/(1-q)$ → (iv) $y = x * x \Rightarrow Y = X^2$.

4b. The DTFT

analysis $X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n}$

synthesis $x[n] = \frac{1}{2\pi}\int_0^{2\pi} X(\Omega)\, e^{j\Omega n}\, d\Omega$

- **Repeats every 2π** in Ω (so the synthesis ∫ can be over any 2π window: 0→2π or −π→π). **Exists if $\Sigma|x[n]| < \infty$**; u[n], $2^n u[n]$ and periodic signals fail.
- **Pairs (|a| < 1):**
  - $a^n u[n] \leftrightarrow 1/(1 - a\cdot e^{-j\Omega})$. **Why:** $a^n e^{-j\Omega n} = q^n$ with $q = a\cdot e^{-j\Omega}$, $|q| = |a|\cdot|e^{-j\Omega}| = |a|$ ($|e^{j\theta}| = 1$ always) → $\Sigma q^n = 1/(1-q)$ needs $|a| < 1$. T02 Q3: $a^{-n}u[n] \rightarrow q = a^{-1}e^{-j\Omega}$, $|q| = 1/|a| < 1 \Leftrightarrow |a| > 1$.
  - $a^{-n}u[-n-1] \leftrightarrow a\cdot e^{j\Omega}/(1 - a\cdot e^{j\Omega})$
  - $(n+1)a^n u[n] \leftrightarrow 1/(1 - a\cdot e^{-j\Omega})^2$ (= x ∗ x)
- **Shift:** $x[n-k] \leftrightarrow e^{-j\Omega k}\cdot X(\Omega)$ (size same, only phase changes).
- **Convolution → product:** $Y(\Omega) = H(\Omega)\cdot X(\Omega)$. H(Ω) = DTFT of h = H(z) at $z = e^{j\Omega}$.
- **One frequency in → same frequency out:** $e^{j\Omega_0 n} \rightarrow H(\Omega_0)\cdot e^{j\Omega_0 n}$. |H| = gain, ∠H = phase shift.

9. Q&A Notes (things I asked about)

- **Period rule:** $\Omega_0 N = 2\pi m$ (N samples, m full laps, both whole); smallest m making N whole. $\Omega_0/2\pi = f/f_s$ = cycles per sample; periodic ⇔ rational. Lowest terms: bottom = N_0, top = cycles.
- **Simplify first:** the rule gives when the *walker* returns; it overcounts only if every sample is 0: $\sin(\pi n) = \sin(3\pi n) = 0$ has period 1 (rule says 2). A sign on Ω or ±2π aliasing doesn't change N_0. $\cos(-\theta) = \cos\theta$; $\sin(-\theta) = -\sin\theta$.
- **$T_s \neq T_0 \neq N_0$:** $T_s = 1/f_s$ (sampler, s); $T_0 = 1/f$ (wave, s); N_0 (samples, whole number). $N_0\cdot T_s = m\cdot T_0$.
- **Causal, formally:** if x = z up to time N, then the outputs are equal up to N. In practice: every x[·] index ≤ n for every n. Traps: x[2n], x[−n]. n·x[n] is fine.
- **Linearity:** $T(a\cdot x_1 + b\cdot x_2) = a\cdot Tx_1 + b\cdot Tx_2$, where a, b scale *values* (x[an] is time scaling, a different thing). **Disprove** with 1 counterexample that switches on the suspect term: for x[n]x[n−1], use v = δ[n] + δ[n−1]: T(v)[1] = 3, T(2v)[1] = 8 ≠ 6. **Prove** with algebra: product terms leave $(a^2-a)pq + (b^2-b)rs + ab(ps + rq)$, via $a^2pq = a\cdot pq + (a^2-a)pq$.
- **Convolution means:** $x = \Sigma x[m]\delta[n-m]$; each kick sends out $x[m]\cdot h[n-m]$; add the copies (LTI only). [2, 1] ∗ [1, 2, 3] = [2, 5, 8, 3].
- **Running sum, h = u** (§2): u in → (n+1)u[n] out. Bounded in, unbounded out → unstable (pole at z = 1).
- **Signal or system?** Transform a signal → X: what it contains. Transform h → H: what the system does to each frequency. Y = H·X. ROC for a signal → which side, whether the DTFT exists; for a system → causal, stable.
- **DFT as matrices:** $\omega = e^{j2\pi/N}$ (N = 4: ω = j), $G[n,k] = \omega^{nk}$; G^H = conjugate transpose (G symmetric → just j ↔ −j). $X = G^H x$ (analysis: row k · x = X[k]); $x = (1/N)\cdot G\cdot X$ (synthesis); works because $G\cdot G^H = N\cdot I$. **Why −j:** the opposite spin stops a matching walker $(j^n(-j)^n = 1$ each time, sums to N); any other frequency cancels to 0.

5. Z-transform and its Pairs

analysis $X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}, \quad z = r\cdot e^{j\Omega}$

synthesis (any r in ROC) $x[n] = \frac{r^n}{2\pi}\int_0^{2\pi} X(re^{j\Omega})\, e^{j\Omega n}\, d\Omega$

- **Meaning:** $X(re^{j\Omega})$ = DTFT of the faded signal $r^{-n}\cdot x[n]$. $|z| = r$ = fading, $\angle z = \Omega$ = frequency. On $|z| = 1$: $X(e^{j\Omega})$ = DTFT.
- **Undo the fading:** if $W(\Omega) = X(re^{j\Omega})$ then $w[n] = r^{-n}x[n]$, so $x[n] = r^n\cdot w[n]$ (= the synthesis formula; in practice invert with §7).
- **Tags:** finite list → $X(z) = \Sigma\, x[n]z^{-n}$. Coefficient of $z^{-k} = x[k]$ (future); of $z^{+k} = x[-k]$ (past).
- **Same thing, two forms:** $z/(z-p) = 1/(1 - p\cdot z^{-1})$.
- **DTFT inside X(z):** the Ω axis from −π to π wraps round the unit circle (Ω = 0 → z = 1, π/2 → j, ±π → −1). DTFT = 1 dial (Ω); Z = 2 dials (r, Ω). Building blocks are spirals $z^n = r^n e^{j\Omega n}$; a pole p means the signal contains p^n. e.g. $2^n u[n]$ faded with r = 4 gives 0.5^n (works for any r > 2, which is the ROC).

Given in the paper (no need to copy): $p^n u[n]$, $-p^n u[-n-1]$, $\rho^n\cos/\sin(\Omega_0 n)u[n]$ pairs; properties (linear, delay, $x[-n]$, $a^n x$, $x*h$); unilateral and bilateral shift rules.

Extra pairs (not given)	X(z)	ROC
$\delta[n-k]$	z^{-k}	all z (not 0 if k>0, not ∞ if k<0)
$p^{n-1}u[n-1]$	$1/(z-p)$	$\|z\| > \|p\|$
$-p^{n-1}u[-n]$ (opt.)	$1/(z-p)$	$\|z\| < \|p\|$
$n\cdot p^n u[n]$	$pz/(z-p)^2$	$\|z\| > \|p\|$
$-n\cdot p^n u[-n-1]$ (opt.)	$pz/(z-p)^2$	$\|z\| < \|p\|$
$(n+1)\cdot p^n u[n]$	$z^2/(z-p)^2$	$\|z\| > \|p\|$

$u[n]$ and $-u[-n-1]$: the given pairs with p = 1. **Ramp via u ∗ u:** $(n+1)u[n] = u * u \leftrightarrow U(z)^2 = z^2/(z-1)^2$; $n\cdot u[n] = (n+1)u[n] - u[n] \leftrightarrow z^2/(z-1)^2 - z/(z-1) = z/(z-1)^2$, $|z| > 1$ (pole on $|z| = 1$ → no DTFT). **Special case $\Omega_0 = \pi/2$:** $p^n\cos(\pi n/2)u[n] \leftrightarrow z^2/(z^2+\rho^2)$; $\rho^n\sin(\pi n/2)u[n] \leftrightarrow \rho z/(z^2+\rho^2)$; both $|z| > \rho$.

6. Poles, Zeros and ROC Rules

- **Poles** = roots of the denominator (×); **zeros** = roots of the numerator (○). First make ONE fraction in +powers of z and cancel: zeros hide in sums $(z/(z-0.6) + z/(z-1.2) = 2z(z-0.9)/(\ldots))$. LCCDE: x-terms → zeros, y-terms → poles.
- **ROC is a ring** $r_1 < |z| < r_2$ with edges at poles; never contains a pole.
- **Which ring:** right-sided → outside the outermost pole · left-sided → inside the innermost pole · two-sided → a ring between poles (can be empty: $2^{|n|}$ needs $|z| > 2$ and $|z| < ½$ → no ROC) · finite length → all z except maybe 0 and ∞.
- **P poles at different radii** → P + 1 possible ROCs, each a different x[n].
- **Which way each pole runs:** pole inside the ring ($|p| \le$ inner edge) → right-sided $p^n u[n]$; pole outside ($|p| \ge$ outer edge) → left-sided $-p^n u[-n-1]$.
- **Stable (DTFT exists)** ⇔ unit circle $|z| = 1$ is in the ROC. A pole on $|z| = 1$ (e.g. z = 1) → no DTFT.
- **Causal** ⇔ ROC is outside the outermost pole **and** includes ∞ (numerator degree ≤ denominator degree in z).
- **Causal and stable** ⇔ every pole strictly inside the unit circle.

7. Inverting X(z)

1. **Factor the denominator** → poles. Mark the ROC's edges.
2. **Numerator has a factor z:** expand X(z)/z = C/(z−p) + D/(z−q) → X(z) = C·z/(z−p) + D·z/(z−q) → p^n-type terms.
3. **No factor z:** expand X(z) = C/(z−p) + D/(z−q) → $p^{n-1}u[n-1]$-type terms.
4. **Cover-up:** C = [(z−p) × the fraction being expanded] at z = p.
5. **Choose each term's side from the ROC** (section 6), then write x[n].
6. **Special denominators:** repeated $(z-p)^2$ → $n\cdot p^n$ or $(n+1)p^n$ pairs · quadratic $z^2 - 2\rho z\cdot\cos\Omega_0 + \rho^2$ → cos/sin pairs (ρ^2 = constant term, then $\cos\Omega_0$ from the middle) · only z^2 appears → set $w = z^2$, do partial fractions in w.

> **Example:** X(z) = z/((z−½)(z−2)), |z| > 2. X/z = 1/((z−½)(z−2)): C = 1/(½−2) = −⅔, D = 1/(2−½) = ⅔. Both poles inside the ring → right-sided: x[n] = [−⅔(½)^n + ⅔·2^n]u[n].

Series expansion (first few samples, right-sided): long-divide in powers of z^{-1}; x[n] = coefficient of z^{-n}. e.g. $(1 + z^{-1})/(1 - ½z^{-1}) = 1 + 1.5z^{-1} + 0.75z^{-2} + 0.375z^{-3} + \ldots \rightarrow x[0..3] = 1, 1.5, 0.75, 0.375$.

Checks: causal x: $x[0] = \lim_{z\to\infty} X(z)$. Plug in n = 0, 1 and compare.

7b. Partial Fractions: the 3 Cases

Expand **X(z)/z** (long-divide first if top degree ≥ bottom), then × z back.

Factor below	Write	Find the numbers
(z−p), distinct	A/(z−p)	cover-up at z = p
$(z-p)^2$, repeated	A/(z−p) + B/$(z-p)^2$	B: cover-up $(z-p)^2$; then A: plug in z = 0
$z^2 + bz + c$ (complex)	(Cz + D)/$(z^2 + bz + c)$	cover-up the others; then compare z^2, const. coeffs

Quadratic → cos/sin pairs: $\rho^2 = c$, $-2\rho\cos\Omega_0 = b$. Split $z(Cz + D) = C\cdot z(z - \rho\cos\Omega_0) + (D + C\rho\cos\Omega_0)\cdot z \rightarrow$ $x = \rho^n[C\cdot\cos(\Omega_0 n) + K\cdot\sin(\Omega_0 n)]u[n]$, $K = (D + C\rho\cos\Omega_0)/(\rho\sin\Omega_0)$. Repeated: $z/(z-p)^2 \leftrightarrow n\cdot p^{n-1}u[n]$ (= extra pair ÷ p).

> **Quadratic e.g.** $X/z = 1/((z-1)(z^2+1)) = A/(z-1) + (Cz+D)/(z^2+1)$. A = 1/(1+1) = ½. 1 = ½$(z^2+1)$ + (Cz+D)(z−1): z^2: 0 = ½ + C → C = −½; const: 1 = ½ − D → D = −½. $z^2 + 1$: ρ = 1, $\Omega_0 = \pi/2$, K = (−½ + 0)/1 = −½. |z| > 1: x = [½ − ½cos(πn/2) − ½sin(πn/2)]u[n] = 0, 0, 1, 1, 0, … ✓

> **Repeated e.g.** $X/z = 1/((z-1)^2(z-2)) = A/(z-1) + B/(z-1)^2 + C/(z-2)$. C = 1/1² = 1; B = 1/(1−2) = −1; z = 0: −½ = −A − 1 − ½ → A = −1. |z| > 2: x = [−1 − n + 2^n]u[n] = 0, 0, 1, 4, … ✓

9. Q&A Notes (continued)

- **Why poles are ROC edges:** pole p ⇔ signal has p^n (×p each step); $\Sigma(p/z)^n$ needs $|z| > |p|$ → edge $|z| = |p|$.
- **1st-order LCCDE:** pole = number before y[n−1] (y = 0.5y[n−1] + x → z/(z−0.5)).
- **Zero:** input z^n → output 0.
- **ROC** = adds up *with* weight z^{-n}; stable = with no weight ($|z| = 1$). A blowing-up h can still have an ROC (Lab 4 Region 0).
- **sin or cos route, same X[k]:** $\sin\theta = \cos(\theta - \pi/2)$, and $1/j = e^{-j\pi/2}$ does the same −π/2 turn. T02: $\sin(2\pi n/6 + \pi/3) = \cos(2\pi n/6 - \pi/6)$ → weights $½e^{\mp j\pi/6}$.

8. Difference Equations (LCCDEs)

No initial conditions (bilateral, LTI) → H(z)

1. Transform every term: $y[n-k] \rightarrow z^{-k}Y$, $x[n-k] \rightarrow z^{-k}X$.
2. $H(z) = Y/X = (b_0 + b_1 z^{-1} + \ldots)/(1 + a_1 z^{-1} + \ldots)$. Multiply top and bottom by the highest power of z to read the poles.
3. One H(z), several ROCs → several systems; use section 6 for causal/stable. Leftover positive power of z on top (from x[n+1]) → not causal for any ROC.
4. $H = z^{-k}\cdot G \Rightarrow h[n] = g[n-k]$: invert G, then shift.

With initial conditions (unilateral, n ≥ 0)

unilateral $Y_+(z) = \sum_{n=0}^{\infty} y[n]\, z^{-n}$

Shift rules $Z_+\{y[n-1]\}$, $Z_+\{y[n-2]\}$ are given in the paper. An input starting at n = 0 has X_+ = its usual right-sided pair.

Recipe

1. Transform both sides with the shift rules.
2. Collect: $Y_+\cdot(1 + a_1 z^{-1} + \ldots) = X_+ -$ (initial-condition terms).
3. **Zero-input:** initial-condition terms ÷ $(1 + a_1 z^{-1} + \ldots)$. **Zero-state:** $H\cdot X_+$.
4. Partial fractions (w = z^2 if only z^2 appears), invert with right-sided pairs (n ≥ 0).
5. **Complete:** $y = y_{zi} + y_{zs}$.
6. **Transient** = terms that die out (system poles inside |z| = 1). **Steady-state** = terms that last (input poles on |z| = 1).
7. **Remove the transient:** set $y[n] = y_{ss}[n]$ for all n; plug into the equation at n = 0 and n = 1; solve for y[−1], y[−2].
8. **Check:** compute y[0], y[1] straight from the equation. Initial value: $y[0] = \lim_{z\to\infty} Y_+(z)$. *(optional) Final value: $y[\infty] = \lim_{z\to 1}(1 - z^{-1})Y_+(z)$, only if all poles inside the unit circle except at most one at z = 1.*

A system is LTI only if its initial conditions are zero.

R. Question-type Recipes (T03–T04)

On the sheet already: 13 Does an ROC exist? *T03 Q1d* → §6 *Which ring* (right part |z| > a, left part |z| < b; none if a ≥ b) · **14 Damped cos** *T03 Q3a* → §7 step 6 + given cos/sin pairs · **15 Inverse** *T03 Q3b–c* → §7 (partial fractions, series) · **17 LCCDE → H, ROCs, h** *T04 Q2* → §8 steps 1–4 + §6.

11 X given on circle r *T03 Q1b*: $W(\Omega) = X(re^{j\Omega})$ = DTFT of $w[n] = r^{-n}x[n]$ → invert W with a §4b pair → $x[n] = r^n\cdot w[n]$. e.g. r = 3, $W = 1/(1 - (j/2)e^{-j\Omega}) \rightarrow w = (j/2)^n u[n] \rightarrow x = (3j/2)^n u[n]$.

12 X(z) and ROC of a signal *T03 Q2*: split into pieces → given pairs (+ §5 extras) (right-sided |z| > |p|, left-sided |z| < |p|) → ROC = overlap of all pieces; finite list → tags (all z except maybe 0/∞) → DTFT exists ⇔ |z| = 1 in ROC.

16 P-Z plot, all ROCs, x for each *T03 Q4*: factor top (○ zeros) and bottom (× poles) → P poles at distinct radii → P+1 ROCs → invert each pole by side (§6) → causal = outside the outermost pole; stable = ring containing |z| = 1; a pole on |z| = r → no ROC includes that radius.

18 LCCDE + ICs, sinusoid in *T04 Q1*: → §8 *Recipe*. No transient ⇔ $y[-1] = y_{ss}[-1]$, $y[-2] = y_{ss}[-2]$; e.g. $y_{ss} = 4\cos(\pi n/2) \rightarrow y[-1] = 0$, $y[-2] = -4$.